Sextuple pairs with square denominators

Problèmes ouverts — A. Dujella · 2026

Problem 3.12 (Dujella, 2026). Are there infinitely many pairs of rational Diophantine sextuples sharing a common triple such that denominators of all elements (in the lowest terms) in the sextuples are perfect squares ?

Remark. In 2024, Rathbun found two pairs of such sextuples. One such pair is

$$\{3/2^2, -12/5^2, -127500/389^2, -2093/50^2, 72512/265^2, -271859133/27970^2\},$$

$$\{3/2^2, -12/5^2, -127500/389^2, 2387/50^2, 435252/685^2, -38178413/6130^2\}.$$